

Astro HW 9

Pierson Lipschultz

November 5, 2025

9.1 Extinction

a) Apparent Mag In front

Given ...

$$m - M = 5 \log(d) - 5$$

$$M = -1.1$$

$$d = 800 \text{pc}$$

We find ...

$$m = 5 \log(800) - 5 - 1.1 \boxed{\approx 8.415}$$

b) Apparent Mag Behind Nebula

Given ...

$$m - M = 5 \log(d) - 5 + A_v$$

$$A_v = 1.1$$

$$d = 820$$

We find that ...

$$m = 5 \log(820) - 5 - 1.1 + 1.1 \boxed{\approx 9.569}$$

c) Error

It would be further than it actually is. This is because the stars' brightness would be dimmer, which would incorrectly imply that the star is further away.

Given ...

$$m = M + 5 \log(d) - 5$$

$$10^{(m-M)/5} = d$$

We find that ...

$$d = 10^{(9.569+1.1+5)/5} \approx 1360.8 \text{pc}$$

$$1360/820 \approx 1.65 \approx \boxed{65\%(\text{error})}$$

9.2 Ionizing stars and HII regions

a) Upper bound of T

Given ...

$$E = h\nu = hc/\lambda$$

$$k = 1.3806 \times 10^{-23}$$

$$h\nu = 13.6J = 13.6\text{ev} \times 1.6021810^{-19}J/\text{ev} = 2.1789 \times 10^{-18}\text{ev}$$

We find that ...

$$2.1789 \times 10^{-18} > 1.3806 \times 10^{-23}T$$

$$\frac{2.1789 \times 10^{-18}}{1.3806 \times 10^{-23}} > T$$

$$\boxed{157827K \gtrsim T}$$

Given ...

b)

Given ...

$$I_\nu = \frac{2h\nu^3}{c^2} e^{-h\nu/k_B T},$$

$$L_\nu = (4\pi R^2)\pi I_\nu = \frac{8\pi^2 h R^2 \nu^3}{c^2} e^{-h\nu/k_B T}.$$

We find that ...

$$Q_\star = \int_{\nu_0}^{\infty} \frac{L_\nu}{h\nu} d\nu = \frac{8\pi^2 R^2}{c^2} \int_{\nu_0}^{\infty} \nu^2 e^{-h\nu/k_B T} d\nu.$$

Let...

$$u = \frac{h\nu}{k_B T}, \quad \nu = \frac{k_B T}{h} u, \quad d\nu = \frac{k_B T}{h} du.$$

Substituting gives...

$$Q_\star = \frac{8\pi^2 R^2}{c^2} \left(\frac{k_B T}{h} \right)^3 \int_{u_0}^{\infty} u^2 e^{-u} du, \quad u_0 = \frac{h\nu_0}{k_B T} = \frac{E_0}{k_B T}.$$

Using the identity...

$$\int u^2 e^{-u} du = -e^{-u}(u^2 + 2u + 2),$$

We find...

$$Q_\star = \frac{8\pi^2 R^2}{c^2} \left(\frac{k_B T}{h} \right)^3 e^{-u_0} (u_0^2 + 2u_0 + 2).$$

Thus...

$$Q_{\star} = \frac{8\pi^2 R^2}{c^2} \left(\frac{k_B T}{h} \right)^3 e^{-E_0/k_B T} \left[\left(\frac{E_0}{k_B T} \right)^2 + 2 \frac{E_0}{k_B T} + 2 \right].$$

Solving for $T = 43650$ and $R = 10.7 R_{\odot} \dots$

$$Q_{\star} = 2.197 \times 10^{49} \text{s}^{-1}$$

c)

Given...

$$R_s = \left(\frac{3}{4\pi} \frac{Q_{\star}}{n_e^2 \alpha(T_e)} \right)^{1/3}$$

$$n_e = 10^9 \text{m}^{-3}, \quad \alpha = 2.6 \times 10^{-19} \text{m}^3 \text{s}^{-1}, \quad Q_{\star} = 2.197 \times 10^{49} \text{s}^{-1}$$

Solving we find that ...

$$R_s \approx 2.7224349 \times 10^{16} \text{m} \approx .882 \text{pc} \approx 88\% \text{ of the normal distance}$$

9.3 Conservation of angular momentum

a) Deriving 17.15

Given ...

$$\omega = 2/5 m r v$$

$$\omega_0 = 2/5 m r_0 v_0$$

$$\omega_f = 2/5 m r_f v_f$$

We find that ...

$$2/5 m r_f v_f = 2/5 m r_0 v_0$$

$$\cancel{2/5 m} r_f v_f = \cancel{2/5 m} r_0 v_0$$

$$v_f = \frac{r_0}{r_f} v_0$$

b) Verify Scaling

Given ...

$$\frac{r_0}{r_f} v_0 = v_f$$

$$\frac{GM}{r_f^2} = \frac{v_f^2}{r_f}$$

We find that...

$$\frac{GM}{r_f} = v_0^2 \frac{r_0^2}{r_f^2}$$

$$\cancel{r_f} \frac{1}{r_f} = v_0^2 \frac{r_0^2}{GM}$$

$$\boxed{r_f = \frac{v_0^2 r_0^2}{GM}}$$

Given...

$$M = 2 \times 10^{30}$$

$$v = 100 \text{ m/s}$$

$$r = 10^{10} \text{ m}$$

$$G = 6.67 \times 10^{-11}$$

We find that...

$$r_f = \frac{100^2 \times 10^{10}}{6.67 \times 10^{-11} \times 2 \times 10^{30}}$$

$$\boxed{r_f \approx 7.496 \times 10^{-15} \text{ m}}$$

c) Enhancement

Given ...

$$\omega_0 = 1/5 m r_0 v_0$$

$$\omega_f = 2/5 m r_f v_f$$

We find that ...

$$2/5 m r_f v_f = 1/5 m r_0 v_0$$

$$\cancel{2} / \cancel{5} m r_f v_f = 1 / \cancel{5} m r_0 v_0$$

$$\boxed{2 \frac{r_0}{r_f} v_0 = v_f}$$

9.4 Post-MS nucleosynthesis

a) Triple-alpha Energy Release

Given ...

$$E = mc^2$$

$$M_{He} = 4.0026032497 \text{ u} \quad M_c = 12 \text{ u}$$

$$u = 1.66053873 \times 10^{-27}$$

We find that ...

$$\Delta M = 12 \text{ u} - 3 \times (M_{He} \times u) \approx -1.29685044 \times 10^{-29}$$

$$E_{rel} = (1.29685044 \times 10^{-29})(300000000)^2$$

$$E_{rel} \approx 1.167165 \times 10^{-12}$$

$$E_{pNuc} = \frac{1.167165 \times 10^{-12}}{12} \approx 9.7263783 \times 10^{-14} \text{ J}$$

b) Star with He Core lifetime

Given ...

$$M_s = 0.3M_\odot = 0.3 \times 1.989 \times 10^{30} \approx 5.97 \times 10^{29} \text{ kg}$$

$$L = 25L_\odot = 25 \times 3.828 \times 10^{26} \approx 9.57 \times 10^{27} \text{ W}$$

$$E_{rxn} = 9.7263783 \times 10^{-14} \text{ J}$$

$$m_{\text{He}} = M_{\text{He}} \times u = 6.64648 \times 10^{-27} \text{ kg}$$

We find that ...

$$N_{\text{He}} = \frac{M_s}{m_{\text{He}}} = 8.98 \times 10^{55}$$

$$N_{\text{rxn}} = \frac{N_{\text{He}}}{3} = 2.99 \times 10^{55}$$

$$E_{\text{tot}} = N_{\text{rxn}} E_{\text{rxn}} = 3.49 \times 10^{43} \text{ J}$$

$$t_{\text{HeBurn}} = \frac{E_{\text{tot}}}{L} = \frac{3.49 \times 10^{43}}{9.57 \times 10^{27}} \approx 3.64 \times 10^{15} \text{ s} \approx 1.16 \times 10^8 \text{ yr}$$

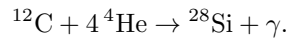
c) Energy released from C → Si and Si → Fe Fusion

Given ...

$$E = mc^2, \quad c = 3.00 \times 10^8 \text{ m s}^{-1}, \quad u = 1.66053873 \times 10^{-27} \text{ kg}$$

$$M_{\text{He}} = 4.0026032497 \text{ u}, \quad M_{\text{C}} = 12.000000 \text{ u}, \quad M_{\text{Si}} = 27.976926 \text{ u}, \quad M_{\text{Fe}} = 55.934942 \text{ u}.$$

(i) $C \rightarrow Si$

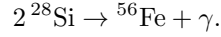


$$\Delta m_{C \rightarrow Si} = (M_{\text{C}} + 4M_{\text{He}}) - M_{\text{Si}} = 0.033486999 \text{ u}.$$

$$E_{C \rightarrow Si} = \Delta m_{C \rightarrow Si} c^2 = (0.033486999 \text{ u}) \left(1.66053873 \times 10^{-27} \frac{\text{kg}}{\text{u}} \right) (3.00 \times 10^8)^2 \approx 5.00 \times 10^{-12} \text{ J}.$$

$$E_n = \frac{E_{C \rightarrow Si}}{28} \approx 1.79 \times 10^{-13} \text{ J}.$$

(ii) $Si \rightarrow Fe$



$$\Delta m_{Si \rightarrow Fe} = (2M_{\text{Si}}) - M_{\text{Fe}} = 0.018910\ \text{u}.$$

$$E_{Si \rightarrow Fe} = \Delta m_{Si \rightarrow Fe} c^2 = (0.018910\ \text{u}) \left(1.66053873 \times 10^{-27} \frac{\text{kg}}{\text{u}} \right) (3.00 \times 10^8)^2 = 2.83 \times 10^{-12}\ \text{J}.$$

$$E_n = \frac{E_{Si \rightarrow Fe}}{56} = 5.05 \times 10^{-14}\ \text{J}$$

As expected, the energy released *per nucleon* decreases as you go from C to Si to Fe.

9.5 Extra Credit

a) Deriving v_{dm}

Given...

$$Ke = 1/2mv^2, \quad GPE = -\frac{GMm}{r}$$

Because the object starts at rest...

$$Ke - GPE_f = -GPE_0$$

We find that ...

$$1/2mv_{dm}^2 - \frac{GMm}{r} = -\frac{GMm}{r_0}$$

$$1/2mv_{dm}^2 = -\frac{GMm}{r_0} + \frac{GMm}{r_0}$$

$$v_{dm}^2 = 2 \left(\frac{GMm}{mr} - \frac{GMm}{mr_0} \right)$$

$$v_{dm}^2 = 2 \left(\frac{GM\cancel{m}}{\cancel{m}r} - \frac{GM\cancel{m}}{\cancel{m}r_0} \right)$$

$$v_{dm} = \sqrt{2GM \left(\frac{1}{r} - \frac{1}{r_0} \right)}$$

Assuming in-fall means negative v...

$$v_{dm} = -\sqrt{2GM \left(\frac{1}{r} - \frac{1}{r_0} \right)}$$

b) Integrating $v(r)$ as dr/dt

Given ...

$$M = (4\pi/3)R_0^3\rho_0$$

We find ...

$$dr/dt = v_{dm} = -\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)}$$

$$dt = \frac{dr}{-\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)}}$$

$$t = \int_{R_0}^r \frac{1}{-\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)}} dr$$

$$t = -\frac{1}{\sqrt{2GM}} \int_{R_0}^r \frac{1}{\sqrt{\left(\frac{1}{r} - \frac{1}{r_0}\right)}} dr$$

$$u = \frac{r}{r_0}, \quad du = 1/r_0 dr, \quad r_0 du = dr, \quad R_0 \rightarrow 0, \quad r \rightarrow 1$$

$$t = -\frac{r_0}{\sqrt{2GM}} \int_0^1 \frac{1}{\sqrt{\left(\frac{1}{ur_0} - \frac{1}{r_0}\right)}} du$$

$$t = -\frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \int_0^1 \frac{1}{\sqrt{\left(\frac{1}{u} - \frac{1}{1}\right)}} du$$

$$t = -\frac{\sqrt{r_0^3}}{\sqrt{2GM}} \int_0^1 \frac{1}{\sqrt{\left(\frac{1-u}{u}\right)}} du$$

$$t = -\frac{\sqrt{r_0^3}}{\sqrt{2GM}} \int_0^1 \sqrt{\left(\frac{u}{1-u}\right)} du, \quad \int_0^1 \sqrt{\left(\frac{x}{1-x}\right)} dx = \frac{\pi}{2}$$

$$t = -\left(\frac{\sqrt{r_0^3}}{\sqrt{2GM}}\right) \left(\frac{\pi}{2}\right)$$